BUSINESS RULES

User Management

1. Every user must have a unique username.
2. Every user must have a password stored as a hashed value.
3. Every user must be assigned exactly one role (Administrator, Lecturer, or Student).
4. A user account may be active or inactive.
5. Administrators can create, update, deactivate, and delete lecturer and student accounts.

Student Management

6. Every student must have a unique matriculation number.
7. Every student belongs to one department.
8. Every student belongs to one programme.
9. A department can have many students.
10. A programme can have many students.
11. A student may register for one or more courses.
12. A course may have many students.

Lecturer Management

13. Every lecturer must have a user account.
14. A lecturer may teach one or more courses.
15. A course is assigned to one lecturer.

Course Management

16. Every course must have a unique course code.
17. A course can have many attendance sessions.
18. Students can only have attendance records for courses they are registered for.

Attendance Management

19. A lecturer can create attendance sessions only for courses assigned to them.
20. Every attendance session belongs to exactly one course.
21. Every attendance session is created by one lecturer.
22. An attendance session may contain many attendance records.
23. Each attendance record belongs to one student and one attendance session.
24. Attendance status must be either "Present" or "Absent".
25. A student can have only one attendance record per session.

Reporting

26. Attendance percentage shall be calculated from attendance records.
27. Students with attendance below the configured threshold shall be marked ineligible.
28. Reports may be generated per student, per course, or per department.

SCHEMA

```
USERS(
    user_id PK,
    username,
    password_hash,
    role,
    status
)

DEPARTMENTS(
    department_id PK,
    department_name
)

PROGRAMMES(
    programme_id PK,
    programme_name,
    department_id FK
)

STUDENTS(
    student_id PK,
    matric_no,
    full_name,
    email,
    programme_id FK,
    user_id FK,
    level
)

LECTURERS(
    lecturer_id PK,
    full_name,
    email,
    user_id FK
)

ADMINISTRATORS(
    admin_id PK,
    full_name,
    email,
    user_id FK
)

COURSES(
    course_id PK,
```

```
        course_code,
        course_title,
        lecturer_id FK
    )
    STUDENT_COURSE(
        student_course_id PK,
        student_id FK,
        course_id FK
    )
    ATTENDANCE_SESSION(
        session_id PK,
        course_id FK,
        lecturer_id FK,
        session_date,
        created_at
    )
    ATTENDANCE_RECORD(
        attendance_id PK,
        session_id FK,
        student_id FK,
        status
    )
```

The database schema is normalized to Third Normal Form (3NF).

• All tables have atomic attributes (1NF).

• No partial dependencies exist (2NF).

• No transitive dependencies exist (3NF).

The design avoids arrays, JSON, JSONB, and other denormalized structures as required by the project specification.